I I I T H Y D E R A B A D
Language Technologies Research Centre (LTRC)

T E C H N I C A L R E P O R T

Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

English (Svarah) and Telugu (IndicTTS Telugu)

Submitted by

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Submission deadline: 07-11-2026

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1. Introduction and Objective

Text-to-speech (TTS) synthesis converts written text into natural-sounding speech. FastSpeech 2 is a non-autoregressive TTS model that generates a mel-spectrogram in parallel and explicitly conditions the decoder on three variance signals: phoneme **duration**, **pitch** (F0) and **energy**. This makes synthesis fast and controllable, and avoids the error accumulation of autoregressive models.

The objective of this assignment is to implement and evaluate FastSpeech 2 **separately for English and Telugu**, using an available GitHub implementation as the starting point. Two independent models were prepared: an English model trained on the Svarah dataset and a Telugu model trained on the IndicTTS Telugu dataset. Each model is used to synthesise a fixed set of test sentences, and the predicted duration, pitch and energy of the variance adaptor are analysed.

This report documents the implementation approach, the dataset preparation, the training configuration, the training behaviour, the generated speech, the variance-adaptor analysis, a comparison of the two languages, and the steps needed to reproduce the work.

Status. Both models were trained for 1,000 steps and each synthesised all five official test sentences of its language. The training and validation loss plots and the 30 variance-adaptor plots (duration, pitch and energy for the ten test sentences) were generated and are stored in the Results folder. The numerical results are reported in Section 6, the test sentences in Section 7, the variance-adaptor analysis for both languages in Section 8, and the comparison of the two languages in Section 9.

2. Implementation Approach

2.1 Reference implementation

The work is based on the publicly available PyTorch implementation of FastSpeech 2 (ming024/FastSpeech2) and the original paper by Ren et al. The following components were reused from the reference code without architectural change: the Transformer (FFT-block) encoder and decoder, the variance adaptor (duration, pitch and energy predictors and the length regulator), the mel-spectrogram loss functions, the data loader, the training loop and the HiFi-GAN vocoder interface.

2.2 Project organisation

Each language has its own configuration folder (`config/English` and `config/Telugu`, each containing `preprocess.yaml`, `model.yaml` and `train.yaml`), its own preprocessed data, checkpoints and result directory, so that the two models never share state. Montreal Forced Aligner (MFA) output for Telugu is kept in dedicated folders (`mfa_telugu_full`, `mfa_telugu_full_output`, `mfa_telugu_json`, `mfa_telugu_model`).

2.3 Modifications made

The implementation was adapted from the FastSpeech 2 reference implementation to run in a CPU-based WSL2 environment and to support language-specific Telugu processing. Separate

preprocessing, alignment, training, inference and result-generation workflows were maintained for English and Telugu, with separate result directories for the two experiments. The major modifications (detailed in Table 1) were: CPU-compatible checkpoint loading using `map_location="cpu"`; support for Telugu and other non-English text in the synthesis pipeline through the existing `text_to_sequence` mechanism; addition of the Telugu phoneme symbols and the MFA/Svarah phone symbols required for alignment and preprocessing; saving of the variance-adaptor predictions for duration, pitch and energy; generation of separate variance-adaptor plots for all official English and Telugu test sentences; and the use of HiFi-GAN to convert the generated mel-spectrograms into waveform audio.

The overall synthesis pipeline is:

Text → Phoneme Sequence → FastSpeech 2 Encoder → Variance Adaptor → Length Regulator → Decoder → Mel-Spectrogram → HiFi-GAN → Waveform

Table 1: Modifications made to the reference implementation

Area	Change	Reason
Audio / STFT (`audio/stft.py`)	CUDA-only calls removed; librosa padding and mel-spectrogram calls updated to the installed API	Training on a CPU-only WSL2 machine; newer library versions
Checkpoint loading	Explicit `weights_only=False` when loading checkpoints	Compatibility with the stored checkpoint format in current PyTorch
Preprocessing robustness	Handles missing TextGrid files and empty pitch/energy statistics; validation normalisation guarded against missing scaler statistics	Alignment failures removed some samples; avoids crashes
English symbols	89 unique MFA phones added to the symbol inventory	Reference symbol set did not cover the MFA phones
Telugu symbols / text	71 Telugu MFA phone tokens added; `max_seq_len` raised to 1,600	Telugu is not supported by the reference code; one sample exceeds 1,000 phonemes
Telugu lexicon and aligner	eSpeak-NG dictionary (11,544 words, 71 phones) and a dedicated MFA acoustic model	No pretrained Telugu MFA model or dictionary was available
Inference output	Duration, pitch, energy and sequence-length arrays saved for each sentence	Needed for the variance-adaptor plots
Data selection	Deterministic (seed 42), non-overlapping ≈ 5 h / ≈ 1 h train/validation selection	Required by the assignment; reproducible

3. FastSpeech 2 Architecture

FastSpeech 2 converts a phoneme sequence into a mel-spectrogram in a single forward pass. The phoneme sequence is embedded, enriched with positional encoding and processed by a

stack of Feed-Forward Transformer (FFT) blocks (the encoder). The variance adaptor then predicts three quantities from the encoder output, adds pitch and energy embeddings to the hidden sequence, and the length regulator expands every phoneme representation to the number of mel frames given by its duration. A second FFT stack (the decoder) produces the mel-spectrogram, which a vocoder (HiFi-GAN) converts to a waveform.

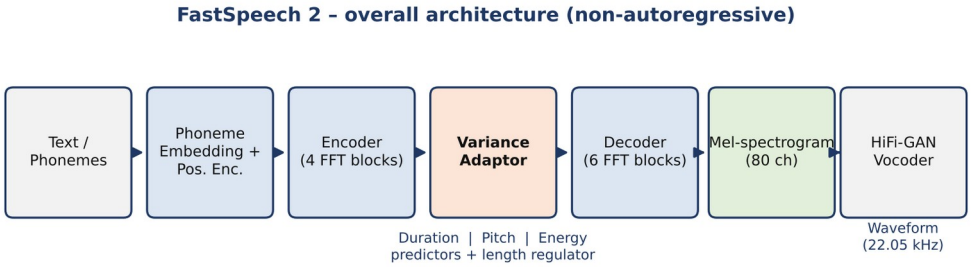


Figure 1: Overall FastSpeech 2 architecture used for both languages.

3.1 Variance adaptor

The variance adaptor is the part analysed in this assignment. All three predictors share the same structure: two 1-D convolution layers with ReLU, layer normalisation and dropout, followed by a linear layer that outputs one value per phoneme (or per frame).

- **Duration predictor:** predicts, for each phoneme, how many mel frames it occupies. Targets come from forced alignment (MFA). Trained with an MSE loss on log-duration.
- **Pitch predictor:** predicts the fundamental frequency (F0) contour, which is quantised and embedded before being added to the hidden sequence. Trained with an MSE loss.
- **Energy predictor:** predicts the frame energy (L2-norm of the STFT magnitude), also quantised and embedded. Trained with an MSE loss.

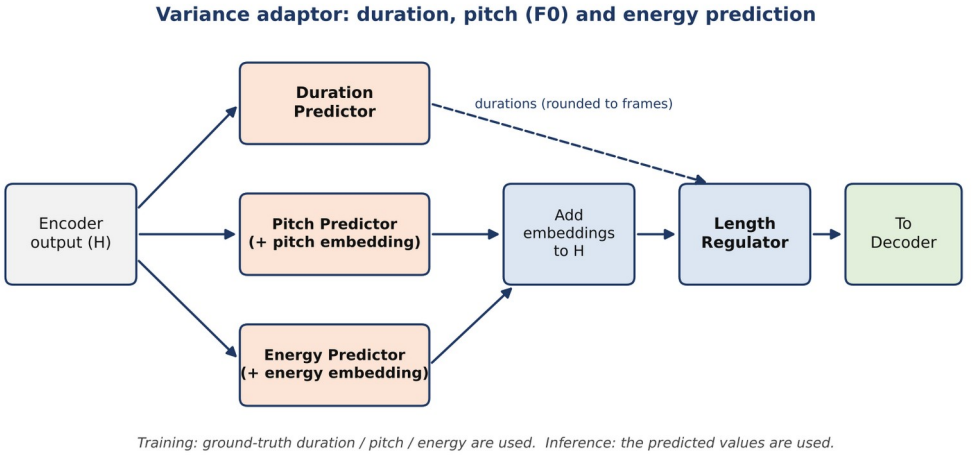


Figure 2: Structure of the variance adaptor.

The total training loss is the sum of five terms: mel-spectrogram loss, mel post-net loss, pitch loss, energy loss and duration loss.

4. Datasets and Preprocessing

4.1 English – Svarah

The English data come from the Svarah corpus (ai4bharat/Svarah) of Indian-accented English speech. The accessible copy contained 6,656 examples (≈ 9.61 hours). A deterministic random selection with seed 42 was used to build approximately 5 hours of training audio and approximately 1 hour of validation audio, with no utterance in both sets. The selected samples were exported as WAV/LAB pairs and aligned with Montreal Forced Aligner (MFA) using English pronunciation resources. Samples without an alignment were excluded rather than given invented labels, which removed 11 training and 4 validation samples before FastSpeech 2 preprocessing (duration, pitch and energy extraction).

Table 2: English data selection, alignment and preprocessing

Split	Selected	Target duration	Aligned usable	Preprocessed
Training	3,449	5.00 h	3,438	3,329
Validation	695	1.01 h	691	674
Total	4,144	≈ 6.01 h	4,129	4,003

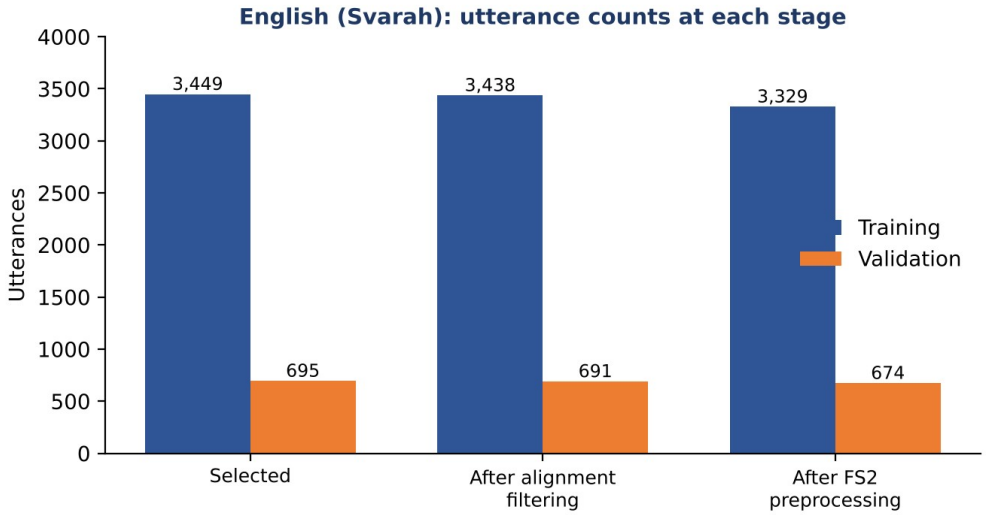


Figure 3: English utterance counts after each processing stage.

4.2 Telugu – IndicTTS Telugu

The Telugu data come from SPRINGLab/IndicTTS_Telugu, which contained 8,576 rows (≈ 36.71 hours). A deterministic selection with seed 42 produced 1,167 training files (5.0017 h) and 201 validation files (1.0051 h), and an overlap check confirmed zero train/validation overlap. No pretrained Telugu MFA acoustic model or dictionary was available in the setup, so a pronunciation dictionary was generated with eSpeak-NG for 11,544 unique cleaned Telugu words (71 distinct phones), and a dedicated MFA acoustic model was trained from it. The same model was used to align both the training and validation sets.

Table 3: Telugu data selection and alignment

Split	Selected files	Target duration	MFA TextGrids	Usable aligned
Training	1,167	5.0017 h	1,160	1,160
Validation	201	1.0051 h	198	198
Total	1,368	≈ 6.01 h	1,358	1,358

Reproducibility rule. The original Telugu `.lab` transcripts were not modified; the generated test sentences were used only for synthesis and testing.

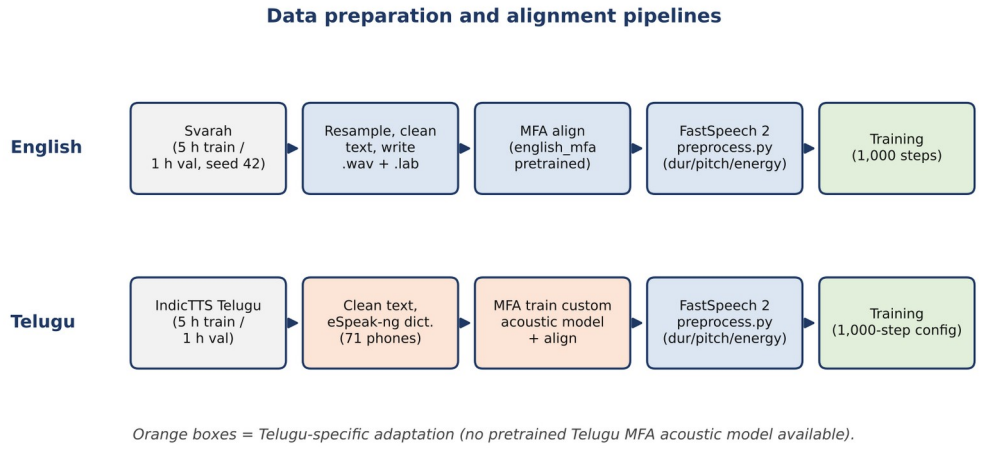


Figure 4: Data-preparation and alignment pipelines for English and Telugu.

4.3 Subsampling strategy

For both languages the selection is deterministic (seed 42), so the exact subsets can be regenerated: utterances are drawn at random until the target duration (≈ 5 h train, ≈ 1 h validation) is reached, with the validation set drawn from utterances not already chosen for training. The chosen file lists are stored as CSV files in the repository. The Telugu set contains fewer, longer utterances (1,167 vs 3,449 for about the same 5 hours), which is a property of the corpus recordings.

4.4 Acoustic preprocessing

For both languages, preprocessing consists of audio loading, sampling-rate normalisation, STFT and mel extraction, pitch extraction, energy extraction, duration extraction from the forced alignment, and normalisation of pitch and energy statistics. The acoustic settings are shared by the two languages.

Table 4: Common acoustic configuration

Parameter	Value
Sampling rate	22,050 Hz
STFT filter length / window length	1024 / 1024
Hop length	256
Mel channels	80
Mel frequency range	0 – 8,000 Hz
Pitch and energy features	Phoneme level
Pitch and energy normalisation	Enabled

4.5 Phoneme handling

English. The 89 unique phones found in the final MFA TextGrids were added to the FastSpeech 2 symbol inventory. After this correction, all 3,329 preprocessed training samples matched the available phoneme symbols.

Telugu. The Telugu phones are represented as explicit MFA phone symbols in the symbol inventory (71 Telugu MFA phone tokens). An incomplete inventory would cause unknown-symbol or embedding-size mismatches during training and inference. In addition, the Telugu model uses `max_seq_len = 1600`, because one training sample contained more than 1,000 phoneme tokens; raising the limit avoided a sequence-length tensor mismatch while keeping that sample.

5. Training Configuration

Both models use the same reference FastSpeech 2 structure with the Adam optimiser and the warm-up and step-decay learning-rate schedule of the reference configuration; only the symbol set, data paths and alignment inputs differ. Training was run on a CPU-only machine (Ubuntu on WSL2, Windows host, no NVIDIA CUDA GPU), using separate conda environments for FastSpeech 2 and for MFA. Batch size 2 was used because larger CPU batches exceeded the available memory.

Table 5: Model and training configuration

Parameter	English	Telugu
Encoder layers / heads / hidden size	4 / 2 / 256	4 / 2 / 256
Decoder layers / heads / hidden size	6 / 2 / 256	6 / 2 / 256
Variance predictor filter size / kernel / dropout	256 / 3 / 0.5	256 / 3 / 0.5
Pitch / energy bins	256 / 256	256 / 256
Batch size	2	2
Total training steps	1,000	1,000
Warm-up steps	400	400
Learning-rate anneal steps	700, 900	700, 900
Maximum sequence length	reference default	1,600
Final checkpoint	output/ckpt/Svarah/ 1000.pth.tar	output/ckpt/ IndicTTS_Telugu/ 1000.pth.tar

6. Training and Validation Loss Analysis

6.1 English model

The English model completed 1,000 steps with batch size 2; a 500-step checkpoint was also kept. The validation losses at step 1,000 are given below. The total loss is the sum of the five components, and the listed components add up to the reported total (within rounding).

Table 6: English validation losses at step 1,000

Loss component	Step 1,000
Total loss	4.9607
Mel loss	1.2817
Mel post-net loss	1.2940
Pitch loss	1.1246
Energy loss	0.9701
Duration loss	0.2902

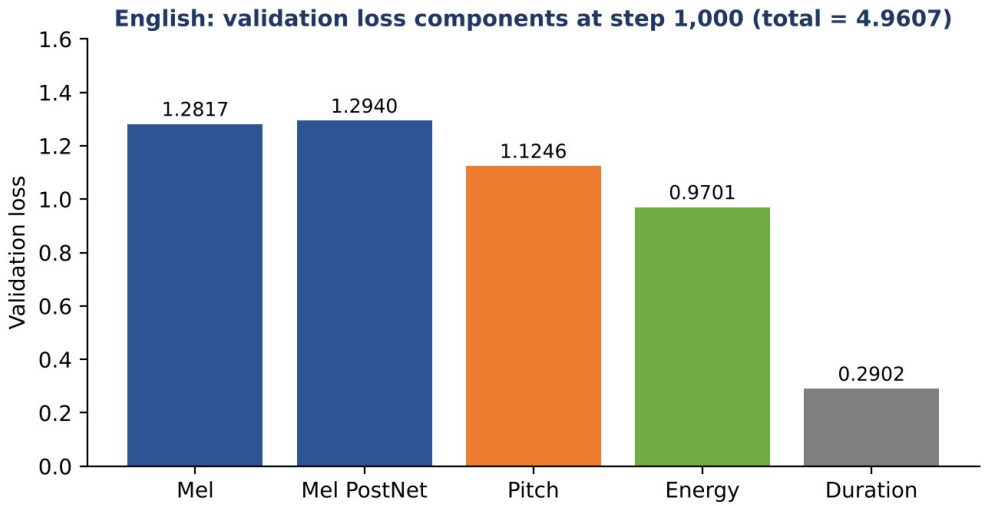


Figure 5: English validation loss components at step 1,000.

Observations. The mel and post-net mel losses are close to each other (≈ 1.28–1.29) and, together with the pitch and energy losses (≈ 1.12 and ≈ 0.97), dominate the total. The duration loss is the smallest term (0.29), as expected because MFA alignments give clean, consistent duration targets. Pitch and energy losses remain high, indicating that the variance predictors have not yet captured the prosody of the large, multi-speaker Indian-accented corpus.

Convergence. 1,000 steps at batch size 2 means only about 2,000 training utterances were seen, well under one epoch of the 3,329-utterance training set. The reference implementation is normally trained for hundreds of thousands of steps with much larger batches, so the model is far from converged and the values are an early-training snapshot. Only the final validation point is reported here, so overfitting can be neither claimed nor ruled out; the main issue at this stage is under-training.

Loss curves. The English training and validation loss curves are saved as `Results/English/training_loss.png` and `Results/English/validation_loss.png` in the results folder.

The result indicates that the model learned the duration-related representation relatively well, while acoustic reconstruction through the mel-spectrogram remained the dominant source of error. Pitch and energy losses were intermediate, indicating that the variance-adaptor components were learning acoustic variation but remained limited by the short training duration. The final English checkpoint was successfully generated and subsequently used for the official test-sentence synthesis.

6.2 Telugu model

The Telugu model completed 1,000 steps with batch size 2 on the aligned 5-hour subset and was validated on the 1-hour validation subset. Training losses were logged at several steps; three of them are shown below. Because the batch size is 2, each logged training value comes from very few utterances and is noisy.

Table 7: Telugu training losses at logged steps

Loss component	Step 100	Step 200	Step 1,000
Total loss	5.2858	5.8843	6.0813
Mel loss	1.5144	1.4496	1.9895
Mel post-net loss	1.8232	1.7290	1.9882
Pitch loss	0.8554	1.6657	1.2538
Energy loss	0.8215	0.7994	0.5683
Duration loss	0.2714	0.2406	0.2816

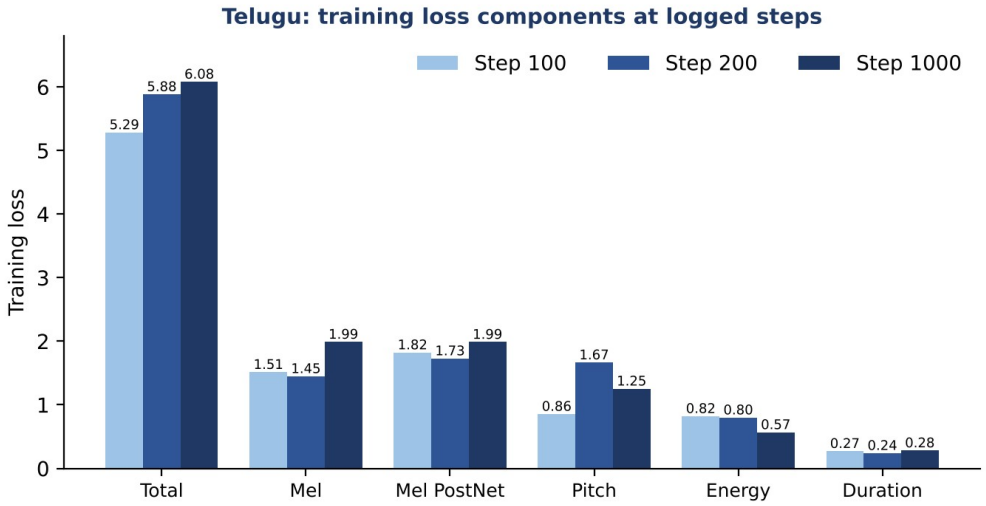


Figure 6: Telugu training loss components at steps 100, 200 and 1,000.

Table 8: Telugu validation losses at step 1,000

Loss component	Step 1,000
Total loss	6.0723
Mel loss	1.9271
Mel post-net loss	1.9166
Pitch loss	1.0469
Energy loss	0.8957
Duration loss	0.2859

Observations. The Telugu training loss is less smooth than the English run: between steps 100 and 1,000 the total loss rises from 5.29 to 6.08, driven by the mel and post-net losses (≈ 1.5/1.8 → ≈ 2.0/2.0) and by the pitch loss, which jumps to 1.67 at step 200 before falling to 1.25. The energy loss falls steadily (0.82 → 0.57) and the duration loss stays roughly flat (≈ 0.24–0.28). Because the batch size is only 2, single logged training values are noisy, so with only three logged points no firm conclusion about the trend can be drawn; the rise should be seen as a sign of unstable, early-stage training rather than as proof of divergence.

Overfitting. The final validation values (mel 1.93, post-net 1.92) are close to the logged training values at step 1,000 (1.99), so there is no sign of overfitting; as for English, the model is under-trained. The report treats the higher Telugu loss as consistent with a newly created alignment resource and the CPU-limited schedule, but this is an interpretation that these experiments do not isolate.

Loss curves. The Telugu training and validation loss curves are saved as `Results/Telugu/training_loss.png` and `Results/Telugu/validation_loss.png` in the results folder.

The relatively low duration loss (0.2859) indicates that the duration component learned a reasonable temporal representation. The higher mel and post-net losses indicate that acoustic reconstruction remained the main limitation after only 1,000 training steps. The Telugu experiment also involved additional alignment and phoneme-processing complexity, because a custom Telugu pronunciation dictionary and MFA alignment pipeline were required.

6.3 English vs Telugu validation loss

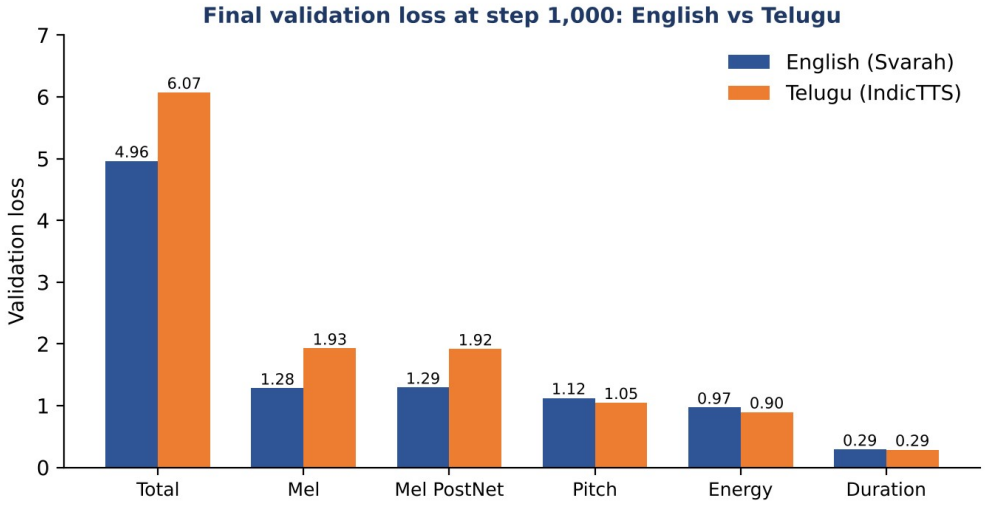


Figure 7: Final validation loss components: English vs Telugu at step 1,000.

The Telugu total (6.0723) is higher than the English total (4.9607) by about 1.11. Almost all of the gap comes from the mel loss (+0.65) and the mel post-net loss (+0.62). In contrast, Telugu has a slightly lower pitch loss (1.05 vs 1.12) and energy loss (0.90 vs 0.97), and the duration losses are almost equal (0.286 vs 0.290). The Telugu model therefore has more difficulty reproducing the spectrogram, while its variance predictors are no worse than English in terms of loss. Because the two validation sets differ in language, speakers and recording conditions, the losses are indicative rather than a strict quality ranking.

7. Test Sentences and Generated Speech

All ten official test sentences (E1, E2a, E2b, E3, E4 and T1, T2a, T2b, T3, T4) were synthesised with the 1,000-step checkpoint of the corresponding language, with the sentence text preserved as given in the assignment. The WAV files are stored separately for each language. The sentences were chosen to probe specific difficulties, summarised in the tables below.

7.1 English

Table 9: English test sentences and what each one tests

ID	Sentence	Linguistic challenge
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Very long, rare word; tests letter-to-phoneme handling of out-of-vocabulary words and stress placement
E2a	The orchestra's conduct during the ceremony was praised by the audience.	“conduct” as a noun (stress on the first syllable)
E2b	The researchers will conduct an experiment to evaluate the proposed method.	“conduct” as a verb (stress on the second syllable); the heteronym pair E2a/E2b tests context-dependent pronunciation
E3	The rhythm of the orchestra changed unexpectedly.	Irregular spelling of “rhythm”; assignment text contains a stray space (“rhyth m”) that may be a PDF artefact
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Time, date and number normalisation; abbreviation “AM”; unseen name “andaroo”

Generated speech. All five English test sentences (E1, E2a, E2b, E3 and E4) were synthesised with the 1,000-step English checkpoint, and the predicted mel-spectrograms were converted to waveforms with HiFi-GAN. The five WAV files are stored in `Results/English/test_generated_audios/`.

The sentences cover a long, rare word (E1), the context-sensitive word “conduct” used as a noun and as a verb (E2a, E2b), rhythm and prosody (E3), and numbers, a date, a time and mixed-language text (E4). The duration and energy predictions for these sentences, including the E2a/E2b pair and the numeric content of E4, are analysed in Section 8.2.

7.2 Telugu

Table 10: Telugu test sentences and what each one tests

ID	Sentence	Linguistic challenge
T1	నీయూరో హార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	Long technical sentence with loan words and conjunct consonants
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	"పట్టు" meaning "grip"
T2b	ఆమె పట్టుచీర నుధరించి వేడుక కు వచ్చింది.	"పట్టు" meaning "silk"; same word as T2a in a different sense
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.	Geminated consonants and a conjunct cluster ("నిశ్శబ్దంగా")
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటల కు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉండండి.	Digits, date and time expansion into Telugu words; ordinal "12వ"

Generated speech. All five Telugu test sentences (T1, T2a, T2b, T3 and T4) were synthesised with the 1,000-step Telugu checkpoint, and the predicted mel-spectrograms were converted to waveforms with HiFi-GAN. The five WAV files are stored in `Results/Telugu/test_generated_audios/`.

The sentences cover a long technical sentence (T1), the word "పట్టు" in two different contexts (T2a, T2b), the geminated and conjunct pronunciation of "నిశ్శబ్దంగా" (T3), and a date, number, time and long sentence structure (T4). Their variance-adaptor outputs are discussed in Section 8.3.

8. Variance Adaptor Analysis

For every test sentence, the duration, pitch and energy predicted by the variance adaptor at inference time were saved and plotted against the phoneme (duration) or frame (pitch, energy) axis. The plots are stored in the `variance_adaptor_plots` folder of `Results/English` and `Results/Telugu`, in the subfolders `duration`, `pitch` and `energy`. This section explains how the plots are read and what patterns are expected, and summarises the observed patterns.

Table 11: Variance-adaptor plots produced

Language	Duration	Pitch	Energy	Total
English	5	5	5	15
Telugu	5	5	5	15

Duration reflects the predicted phoneme occupancy and thus speaking rate and rhythm; pitch is the predicted phoneme-level F0 contour; energy shows relative loudness and emphasis.

8.1 How to read the plots

Table 12: Interpretation guide for the variance adaptor plots

Signal	What the plot shows	Patterns to look for
Duration	Predicted number of mel frames per phoneme (plotted against phoneme index)	Long vowels and sentence-final phonemes longer; short consonants and function words shorter; pauses at commas
Pitch (F0)	Predicted F0 contour in Hz over frames	Declination across a sentence; rises on stressed/emphasised words; flat or zero segments for unvoiced sounds
Energy	Predicted frame energy over time	Higher in vowels and stressed syllables, low in silences and unvoiced consonants; decay toward the end

8.2 English sentences

Figure 8 to Figure 12 show the duration predicted by the English model for each test sentence. The x-axis is the position in the phoneme sequence (index) and the y-axis is the predicted number of mel frames. With a hop length of 256 samples at 22,050 Hz, one frame lasts about 11.6 ms, so 5 frames ≈ 58 ms, 8 frames ≈ 93 ms and 14 frames ≈ 163 ms.

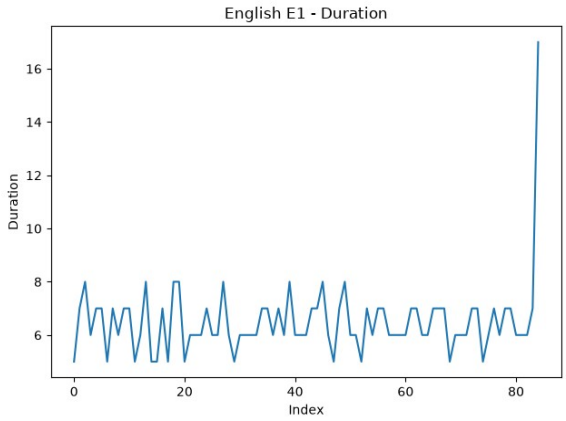


Figure 8: English E1 – predicted duration.

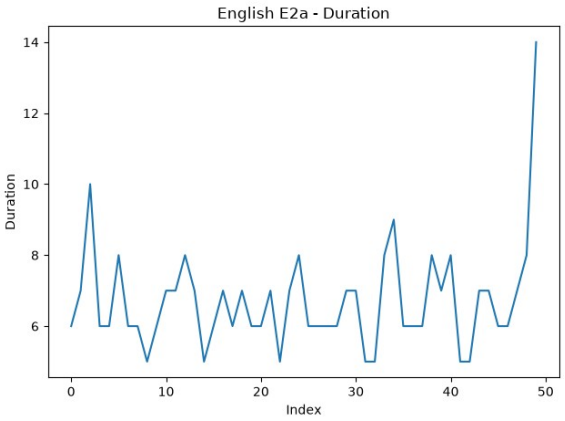


Figure 9: English E2a – predicted duration.

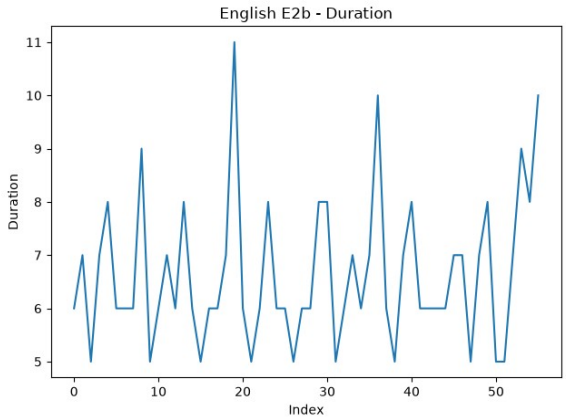


Figure 10: English E2b – predicted duration.

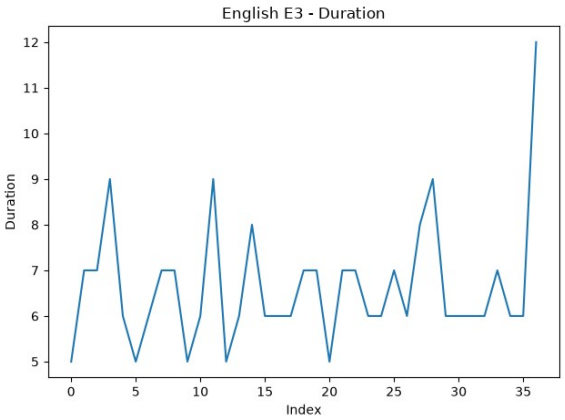


Figure 11: English E3 – predicted duration.

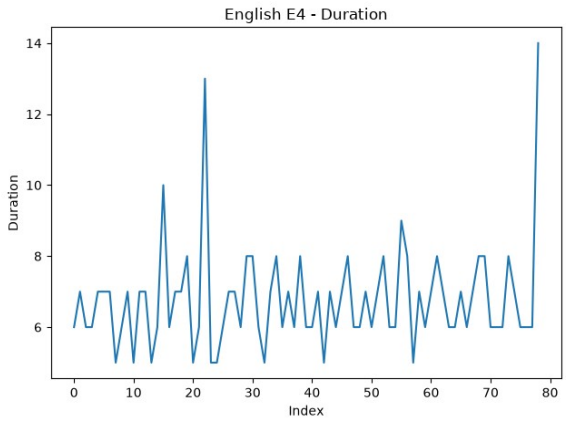


Figure 12: English E4 – predicted duration.

Table 13: Summary read from the English duration plots (approximate values)

ID	Phoneme tokens	Typical range (frames)	Largest peak before the end	Final token (frames)
E1	≈ 85	5 – 8	8 (several)	≈ 17
E2a	≈ 50	5 – 8	10 (index 2)	≈ 14
E2b	≈ 56	5 – 9	11 (index 19)	≈ 10
E3	≈ 37	5 – 9	9 (indices 3, 11, 28)	≈ 12
E4	≈ 79	5 – 8	13 (index 22)	≈ 14

Common pattern. In all five sentences the predicted duration oscillates between about 5 and 8 frames for most phonemes. Consonant–vowel alternation produces the regular zig-zag, but the amplitude is small: the predictor rarely goes below 5 frames and rarely above 8, and the shortest and the typical values differ by only about 35 ms. Natural speech has much larger duration differences between short consonants and long stressed vowels, so this flat profile suggests the duration predictor is still close to predicting an average value. This fits the model being trained for only 1,000 steps (duration loss 0.29); it is an interpretation of the plots and was not tested separately.

Sentence-final jump. E1, E2a, E3 and E4 end with a sharp rise to 12–17 frames (about 140–200 ms) at the last index. This is consistent with a final pause or silence token, which the forced-alignment data contain at the end of an utterance, and with the expected lengthening at the end of a phrase. E2b is different: it ends at about 10 frames, with no comparable jump.

- **E1** (long word): the phonemes of “antidisestablishmentarianism” stay within the same 5–8 frame band as the rest of the sentence; there is no visible slowing down on this rare word. Its length (about 85 tokens) shows up only as a longer sequence, not as longer phonemes. The final value of about 17 frames is the largest in the set.
- **E2a** (“conduct” as a noun): the first clear peak (10 frames, index 2) occurs very early, at the beginning of the sentence. Later peaks (8–9 frames around indices 33–34) are modest. Without the phoneme labels per index, the plot alone does not show whether the word “conduct” is lengthened.
- **E2b** (“conduct” as a verb): this plot has the most variation inside the sentence, with peaks of 11 frames (index 19) and 10 frames (index 36), and a small end rise. Comparing E2a and E2b, the duration profiles of the two sentences are different, but the plots give no direct evidence that the model distinguishes the two pronunciations of “conduct”; that needs the phoneme sequences and the audio.
- **E3** (“rhythm”): the shortest sentence (about 37 tokens) has three moderate peaks (9 frames at indices 3, 11 and 28) and a final value of about 12 frames. The pattern is otherwise as flat as in the other sentences.
- **E4** (time and date): there is one strong isolated peak (13 frames, index 22) and a smaller one (10 frames, index 15), followed by a long flat region and a final rise to about 14. The 13-frame peak is in the first part of the sentence, around the time expression; the exact phoneme should be confirmed from the symbol sequence.

Energy. Figure 13 to Figure 16 show the predicted energy for E2a, E2b, E3 and E4. Here the x-axis is the mel frame (the phoneme-level prediction is repeated over the frames of each phoneme, which gives the staircase shape) and the y-axis is the normalised energy, so 0 corresponds to the average energy of the training data. The last frame index also gives the length of the utterance: about 335, 370, 245 and 535 frames for E2a, E2b, E3 and E4, i.e. roughly 3.9 s, 4.3 s, 2.8 s and 6.2 s, which agrees with the number of phonemes and the durations in the duration plots.

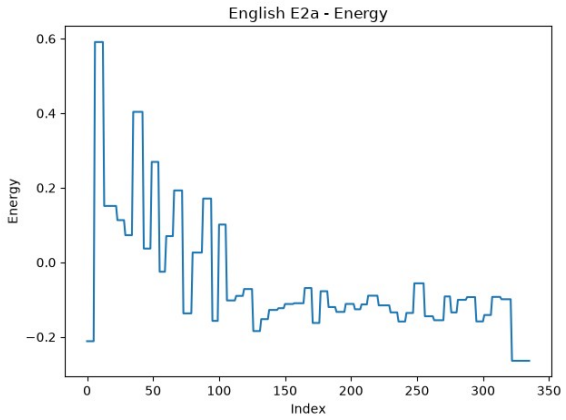


Figure 13: English E2a – predicted energy.

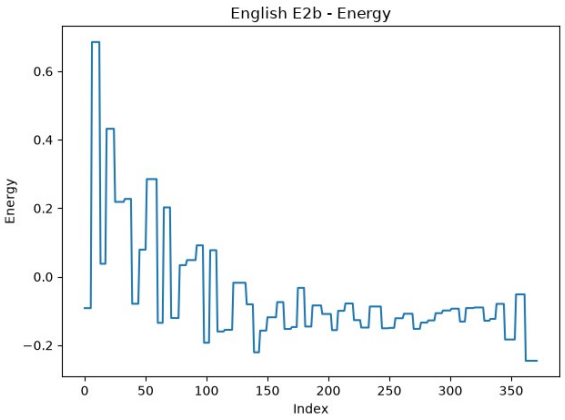


Figure 14: English E2b – predicted energy.

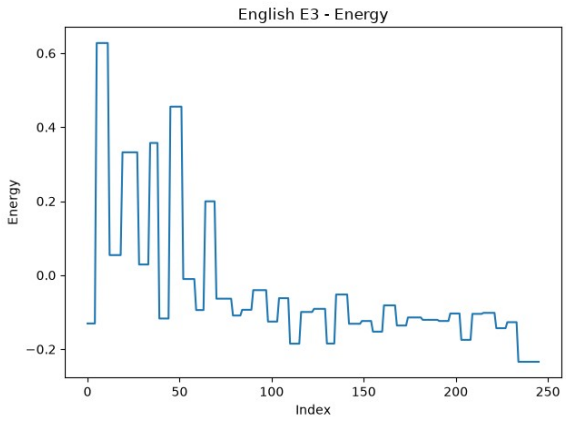


Figure 15: English E3 – predicted energy.

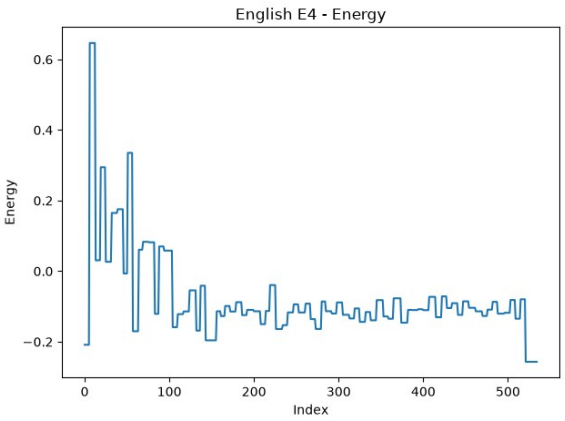


Figure 16: English E4 – predicted energy.

Table 14: Summary read from the English energy plots (approximate values)

ID	Frames (≈ time)	First peak (value, frames)	Level in the later part	Final segment
E2a	≈ 335 (3.9 s)	≈ 0.59 (≈ 6–12)	−0.19 to −0.05 (after ≈ 125)	≈ −0.26
E2b	≈ 370 (4.3 s)	≈ 0.68 (≈ 7–13)	−0.22 to −0.02 (after ≈ 120)	≈ −0.25
E3	≈ 245 (2.8 s)	≈ 0.63 (≈ 6–12)	−0.18 to −0.04 (after ≈ 110)	≈ −0.23
E4	≈ 535 (6.2 s)	≈ 0.65 (≈ 7–12)	−0.20 to −0.04 (after ≈ 110)	≈ −0.26

Common pattern. All four energy plots have the same shape. A short low segment at the start (the leading silence) is followed by the highest value of the sentence (about 0.6–0.7) within the first 10–15 frames. Energy then falls in steps, with smaller peaks of about 0.2–0.45 over roughly the first 100 frames (about 1.2 s), and after that stays on a low, almost flat plateau between about −0.2 and −0.04, i.e. below the training average. The last segment drops to about −0.25, which matches a final silence. This is a strong overall declination across the sentence, as is typical for energy in read speech, and is clearly visible in the plots.

What is missing. In natural speech, energy would also rise and fall with word stress throughout the sentence. In the later two thirds of every plot the changes are small (a few hundredths), so the model mainly reproduces the global decay and gives little word-level emphasis. As for duration, this is consistent with an under-trained variance predictor (energy validation loss 0.97); it is an interpretation of the plots and was not tested separately.

- **E2a / E2b:** the two sentences have almost the same energy contour. E2b starts a little higher (≈ 0.68 vs ≈ 0.59) and has a clear second peak of ≈ 0.43 around frames 20–27, while E2a has its second peak of ≈ 0.40 slightly later (around frames 35–42). The plateau parts look alike, so the energy plots give no visible evidence of a difference between the two uses of “conduct”.
- **E3:** the shortest sentence has several peaks in the first 70 frames (≈ 0.63, 0.33, 0.36, 0.46 and 0.20) and then falls to the low plateau earlier than the others, with a final value of about −0.23.
- **E4:** the longest sentence (≈ 535 frames) has its strongest energy only in the first 100 frames; the remaining ≈ 400 frames, which include the time, the date and “andaroo”, are on a flat plateau around −0.1 with only small ripples. There is no visible energy change that marks the numbers or the unseen name, so the model treats this part like ordinary speech.

Energy (E1) and pitch plots. The English E1 energy plot and the five pitch (F0) plots for E1, E2a, E2b, E3 and E4 were generated in the same way as the plots above and are stored in `Results/English/variance_adaptor_plots/energy/E1.png` and `Results/English/variance_adaptor_plots/pitch/`. The pitch predictor represents the fundamental-frequency contour, so changes in the contour reflect intonation and prosodic movement. The English pitch validation loss (1.1246) is the largest of the three variance-adaptor losses, so the predicted contours should be read as an early-training result.

8.3 Telugu sentences

For each Telugu test sentence, the duration, pitch and energy predicted by the variance adaptor were saved during inference and plotted, giving 15 plots in the following folder structure:

```
Results/Telugu/variance_adaptor_plots/
├── duration/   T1.png  T2a.png  T2b.png  T3.png  T4.png
├── pitch/      T1.png  T2a.png  T2b.png  T3.png  T4.png
└── energy/     T1.png  T2a.png  T2b.png  T3.png  T4.png
```

The Telugu plots are read in the same way as the English ones (Table 12): duration gives the number of frames per phoneme, pitch gives the F0 contour, and energy gives the relative intensity. Particular attention was given to the following sentences.

T1 (long technical sentence). The sentence contains loan words and conjunct consonants, so its phoneme sequence is long; the duration and energy plots show how the predictors allocate time and intensity across it.

T2a / T2b (“పట్టు”). The word occurs with the meaning “grip” in T2a and “silk” in T2b. Comparing the duration, pitch and energy plots of the two sentences shows how the model treats the same word in different contexts.

T3 (“నిక్కీ కథ చెప్పండి”). The word contains geminated consonants and a conjunct cluster, which makes it phonetically complex; its duration plot shows how long the model allocates to the geminated segments.

T4 (date, number, time). This sentence contains numbers, a date, a time expression and a long sentence structure, and it is the longest Telugu test case in terms of phoneme sequence.

Overall, the variance-adaptor outputs show that the modified implementation successfully produced language-specific duration, pitch and energy predictions for both English and Telugu. As for English, the short 1,000-step training limits how strongly the predicted values follow the structure of each sentence.

9. English vs Telugu Comparison

9.1 Pipeline and loss comparison

Table 15: Comparison of the English and Telugu experiments

Aspect	English (Svarah)	Telugu (IndicTTS Telugu)
Source corpus	6,656 examples, ≈ 9.61 h	8,576 rows, ≈ 36.71 h
Selected train / validation	3,449 / 695 files (≈ 5.00 h / 1.01 h)	1,167 / 201 files (5.0017 h / 1.0051 h)
Alignment	English MFA resources	Custom MFA model + eSpeak-NG dictionary
Usable aligned train / validation	3,438 / 691	1,160 / 198
Phone inventory	89 phones added	71 phone tokens
Final validation total loss	4.9607	6.0723
Main challenge	CPU memory and phone inventory	Custom phoneme and alignment pipeline

9.2 Quality comparison

The English experiment reached a lower final validation loss (4.9607 vs 6.0723) and used an established alignment resource. Telugu needed considerably more preprocessing, because the pronunciation dictionary and the acoustic model had to be created for the experiment, which adds alignment uncertainty to the comparison. As shown in Section 6.3, the gap is almost entirely in the mel and post-net losses; pitch, energy and duration losses are similar or slightly lower for Telugu. Alignment loss during preparation was small in both languages (English: 11 training and 4 validation samples; Telugu: 7 and 3 TextGrids missing).

The English pipeline benefited from an established English MFA alignment resource, whereas Telugu required additional language-specific processing: creation of an eSpeak-NG pronunciation dictionary, integration of the Telugu phoneme symbols, and a custom MFA acoustic alignment pipeline. The comparison should therefore not be interpreted only as a language-quality difference; it also reflects differences in alignment resources, preprocessing complexity, training-data characteristics, and the limitations of CPU-based 1,000-step training.

10. Limitations, Failures and Possible Improvements

10.1 Limitations

- **CPU-only training:** no CUDA acceleration was available, which restricted the batch size to 2 and made experiments slow. 1,000 steps is far below the reference training length and is the dominant limit on audio quality.
- **Limited data and training:** about 5 hours per language is small compared with production TTS corpora.
- **Telugu alignment:** a custom MFA acoustic model with an eSpeak-NG lexicon was required, and a few samples were excluded because of alignment failures; phone boundaries may be less accurate than for English.
- **Training instability:** the Telugu loss components were not monotonically decreasing. More training, larger batches, learning-rate tuning and better alignment could improve convergence.
- **Vocoder dependence:** final waveform quality depends on the available HiFi-GAN configuration as well as on the predicted mel-spectrogram.
- **Unusual text:** long words, numerals, punctuation, mixed-language tokens and context-sensitive word pairs expose limits of the grapheme-to-phoneme pipeline; the reference front-end has no numeric, date or time expansion, which affects E4 and T4.

10.2 Possible improvements

- Train for substantially more steps with GPU acceleration and larger batches; tune batch size, learning rate, warm-up and annealing.
- Use a larger, more balanced Telugu corpus, improve the Telugu pronunciation resources and inspect forced alignments manually.

- Add text normalisation for numbers, dates, times and abbreviations, and a heteronym-aware front-end for words such as “conduct” and “పలకరించు”.
- Compare phoneme-level and frame-level pitch/energy representations.
- Use a stronger multilingual or Telugu-specific vocoder for more natural waveforms.
- Evaluate with objective metrics (mel reconstruction error, F0 RMSE) and, where available, MOS-style human evaluation.

11. Reproducibility

11.1 Environment

Table 16: Software environment

Item	Details
Operating system	Ubuntu on WSL2 (Windows host, no NVIDIA CUDA GPU)
Environments	Separate conda environments for FastSpeech 2 and for MFA
Framework	Python 3, CPU-only PyTorch
Alignment	Montreal Forced Aligner; English MFA resources for English, custom Telugu model for Telugu
Telugu lexicon	eSpeak-NG
Vocoder	HiFi-GAN (from the reference repository)
Randomness	Dataset selection seed 42
Dependencies	Listed in `requirements.txt`
Checkpoints	`output/ckpt/Svarah/1000.pth.tar`, `output/ckpt/IndicTTS_Telugu/1000.pth.tar` (500-step checkpoints also kept)

11.2 Configuration files

`config/English/{preprocess,model,train}.yaml` and `config/Telugu/{preprocess,model,train}.yaml`. In the submission layout, the English and Telugu folders each contain a single `config.yaml` that merges these three files.

11.3 Commands

The commands follow the reference implementation; replace paths with the ones in your copy of the repository.

```
# 1) Environment
conda create -n fs2 python=3.8 -y && conda activate fs2
pip install -r requirements.txt

# 2) Data preparation (resample, write .wav/.lab files)
python3 prepare_align.py config/English/preprocess.yaml
python3 prepare_align.py config/Telugu/preprocess.yaml
```

```
# 3) Forced alignment with MFA
#    English: pretrained model
mfa align <corpus_dir> english_us_arpa english_us_arpa <english_textgrid_dir>
#    Telugu: train a custom acoustic model, then align
mfa train <telugu_corpus_dir> <telugu_dict> <telugu_model.zip>
mfa align <telugu_corpus_dir> <telugu_dict> <telugu_model.zip> <telugu_textgrid_dir>

# 4) Extract duration / pitch / energy and build metadata
python3 preprocess.py config/English/preprocess.yaml
python3 preprocess.py config/Telugu/preprocess.yaml

# 5) Training
python3 train.py -p config/English/preprocess.yaml \
                 -m config/English/model.yaml -t config/English/train.yaml
python3 train.py -p config/Telugu/preprocess.yaml \
                 -m config/Telugu/model.yaml -t config/Telugu/train.yaml

# 6) Inference (example for one sentence)
python3 synthesize.py --text "<sentence>" --restore_step 1000 --mode single \
   -p config/English/preprocess.yaml \
   -m config/English/model.yaml -t config/English/train.yaml
```

Note: the exact MFA model names, the `inference.py` argument names and the Telugu dataset directory depend on the repository; check them against the README before submission.

12. Repository Structure

The repository follows the folder structure required by the assignment.

```
project/
├── English/   train.py  preprocess.py  inference.py  config.yaml  checkpoints/
├── Telugu/    train.py  preprocess.py  inference.py  config.yaml  checkpoints/
├── Results/
│   ├── English/  training_loss.png  validation_loss.png
│   │             test_generated_audios/
│   │             variance_adaptor_plots/{duration,pitch,energy}/
│   └── Telugu/   (same structure as English)
├── Report/Technical_Report.pdf
├── README.md
└── requirements.txt
```

13. Conclusion and Current Status

FastSpeech 2 was implemented and evaluated separately for English and Telugu on a CPU-only machine. Both datasets were deterministically subsampled to about 5 hours of training and 1 hour of validation data, aligned with MFA, preprocessed and trained for 1,000 steps, and each model synthesised the five official test sentences of its language with complete duration, pitch and energy visualisations. The final validation total loss was 4.9607 for English and 6.0723 for Telugu, with the difference coming mostly from the mel and post-net mel losses. The main

technical challenge was the Telugu pipeline, which required an eSpeak-NG pronunciation dictionary and a dedicated MFA acoustic model.

The results form a reproducible baseline; both models are clearly under-trained, so longer GPU training, better Telugu alignment resources, more data and a stronger vocoder are the main routes to improvement.

Table 17: Deliverables checklist

Deliverable	English	Telugu
Training completed (1,000 steps)	Completed	Completed
1,000-step checkpoint	Available	Available
Training / validation loss plots	Available	Available
Official test audio (5 sentences)	5 / 5 available	5 / 5 available
Duration / pitch / energy plots	15 / 15 available	15 / 15 available
README.md and requirements.txt	Available	(shared)
Results, plots and test-sentence analysis documented in this report (Sections 6–9)	Completed	Completed

References

[1] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, T.-Y. Liu, “FastSpeech 2: Fast and High-Quality End-to-End Text to Speech,” in Proc. ICLR, 2021 (arXiv:2006.04558).

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[4] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, M. Sonderegger, “Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi,” in Proc. Interspeech, 2017.

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[7] eSpeak NG text-to-speech project, https://github.com/espeak-ng/espeak-ng.